



Final End-to-End Project Documentation
Apr 2025 – Mar 2026

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PHASE 1 — Data Collection & Extraction

Data Source

- Official TRAI website — www.trai.gov.in
- 12 monthly PDF press releases downloaded, April 2025 to March 2026
- Each PDF ran 17–25 pages, covering subscriber telemetry, operator breakdowns, broadband segments, MNP data, 5G FWA annexures, VLR counts and M2M data

Extraction Process

- Claude used as a semantic parser — raw PDF text pasted in and converted to clean CSV rows
- Gemini used for gap-filling — May, July and September 2025 were missing from the TRAI archives
- Gemini also validated column names and the mathematical relationships across all 12 months

Structuring the 59 Columns

Before any SQL was written, the 59 raw KPI columns were sorted into logical business categories using an online drag-and-drop mind-map tool — a quicker way to visually group columns than working it out on paper. This sketch became the blueprint for the 12 BigQuery views built in Phase 2.

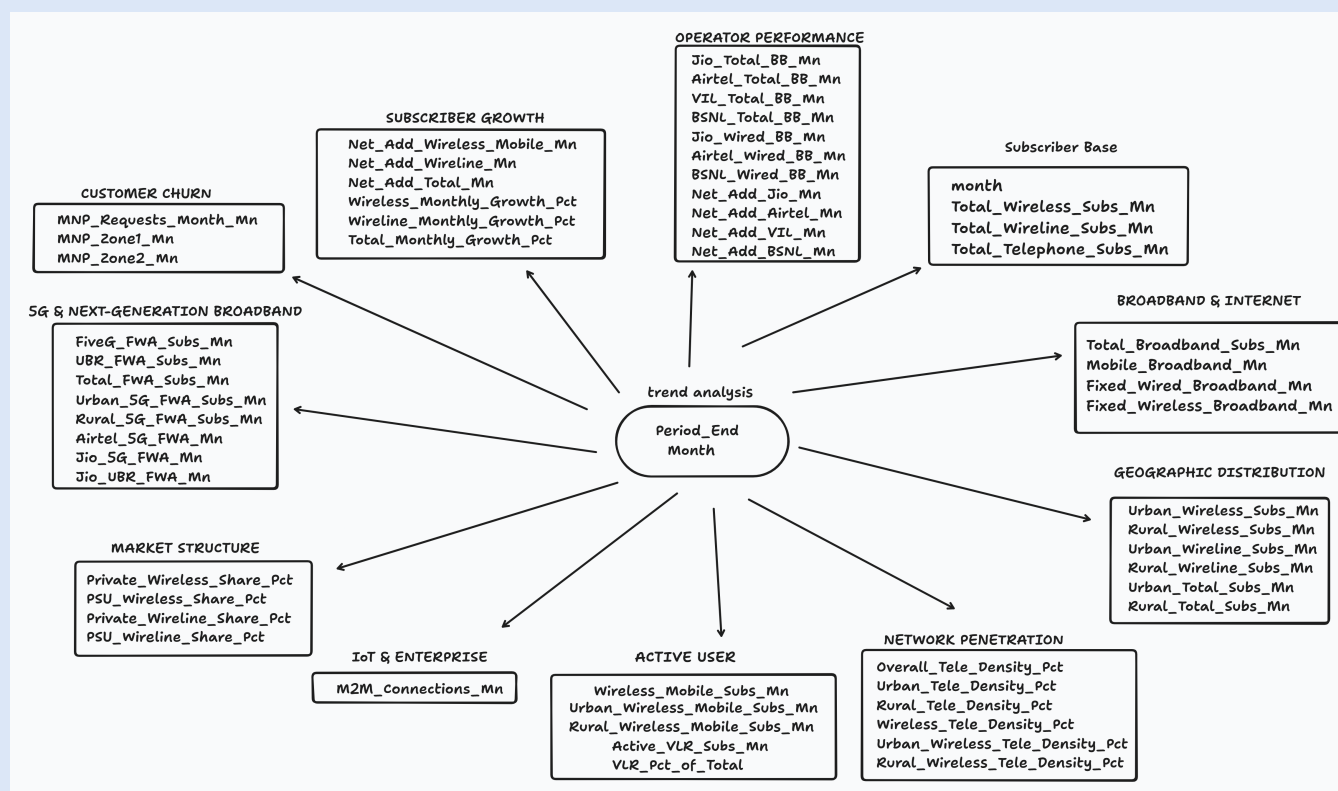


Fig — Rough sketch used to segregate the 59 KPI columns into logical categories (Subscriber Base, Broadband, Geographic, Tele-Density, Churn, FWA/5G, Market Structure, etc.) before building the BigQuery views.

Final Master Dataset

Property	Detail
File Name	THEFINAL_DATASET.xlsx
Rows	12 (one per calendar month)
Columns	59 KPI columns
Period	April 2025 – March 2026
Operators	Jio, Airtel, Vodafone Idea (Vi), BSNL, MTNL (PSU aggregate)

PHASE 2 — SQL Analysis in Google BigQuery

What Was Done

- Dataset loaded into a dedicated BigQuery project
- 12 domain-specific views created to organise the data by business category
- 12 analytical SQL queries written — one per business question
- Every query verified with results, and insights documented

12 Views Created

#	View Name	Key Columns
1	REPORTING_PERIOD	Period_End, Month
2	SUBSCRIBER_BASE_MARKET_SIZE	Wireless, Wireline, Total subscribers
3	BROADBAND_INTERNET_ADOPTION	Mobile, Fixed Wired, Fixed Wireless, Total BB
4	GEOGRAPHIC_DISTRIBUTION	Urban & Rural subscriber splits
5	TELE_DENSITY_DIGITAL_INCLUSION	Overall, Urban, Rural tele-density %
6	ACTIVE_SUBSCRIBERS	VLR active subs, VLR % of total
7	FWA_AND_5G_BROADBAND	5G FWA, UBR FWA, Urban/Rural FWA, Operator FWA
8	IOT_ENTERPRISE_CONNECTIVITY	M2M connections
9	MNP_AND_CHURN	MNP requests, Zone-1, Zone-2
10	SUBSCRIBER_GROWTH	Net additions, Monthly growth %
11	OPERATOR_PERFORMANCE	Jio, Airtel, Vi, BSNL broadband & net adds
12	MARKET_SHARE_COMPETITION	Private vs PSU wireless & wireline share %

12 Business Questions Answered in SQL

Q#	Topic	What Was Answered
Q1	Competitor Battle	Ranked Jio, Airtel, Vi, BSNL by 12-month cumulative net subscriber gain
Q2	5G FWA Adoption	Month-over-month growth rate of 5G FWA — accelerating vs plateauing
Q3	Urban vs Rural Divide	Gap between urban and rural tele-density across 12 months
Q4	MNP Churn Alert	Zone-1 vs Zone-2 churn comparison and spike identification
Q5	Network Health	VLR active SIM ratio (VLR %) per month — best and worst months
Q6	Wireline Turnaround	Wireless vs wireline monthly growth — is landline recovering?
Q7	M2M Automation	Average M2M growth rate and 12-month future projection
Q8	Private vs PSU	Market share split across wireless and wireline segments
Q9	Broadband Drivers	Mobile vs Fixed Wired vs FWA share of total broadband
Q10	Volatility Check	Standard deviation of net additions — steady or erratic growth?
Q11	Jio vs Airtel BB	Subscriber gap trend — is Airtel closing in on Jio?
Q12	FWA Urban vs Rural	Urban/rural share of 5G FWA — rural adoption growing?

PHASE 3 — Microsoft Fabric Setup

Fabric Workspace

Workspace name: **Telecom_project**

Two Methods Used to Load Data

Item	Lakehouse Method	Warehouse Method (via Dataflow Gen2)
Item Created	Telecom_lakehouse	Telecom_Warehouse
How Data Loaded	Direct file upload into Lakehouse	Dataflow Gen2 — Source: Lakehouse → Dest: Warehouse
Key Transform	None (raw load)	Date column changed from Text to Date type
SQL Access	SQL Analytics Endpoint (read-only)	Full T-SQL — CREATE VIEW supported
Status	Completed	Completed

Why Warehouse Over the Lakehouse SQL Endpoint

The Lakehouse landed the raw dataset, but its SQL Analytics Endpoint is read-only and does not support **CREATE VIEW**. Because of account-level limitations, the dataset couldn't be dumped directly into the Warehouse either — so a **Dataflow Gen2** was used to move the data from the Lakehouse (source) into the Warehouse (destination). The Warehouse supports full T-SQL (CREATE VIEW, INSERT, UPDATE) and connects directly to Power BI Service in the cloud, with no Desktop install needed — the standard enterprise pattern of Table → Views → Power BI.

12 Views Recreated in the Fabric Warehouse

- All 12 BigQuery views converted from BigQuery SQL to Fabric T-SQL syntax
- Key change: backtick references replaced with [dbo].[telecom] — Fabric is case-sensitive, so this was kept lowercase
- CREATE OR REPLACE VIEW changed to CREATE VIEW
- All 12 views created successfully in Telecom_Warehouse

Key Syntax Differences — BigQuery vs Fabric T-SQL

BigQuery	Fabric Warehouse (T-SQL)
CREATE OR REPLACE VIEW `project.dataset.ViewName`	CREATE VIEW [dbo].[ViewName]
FROM `project.dataset.Telecom`	FROM [dbo].[telecom]
STDDEV()	STDEV()
CONCAT(value, '%')	CONCAT(CAST(value AS VARCHAR), '%')
Backtick for special-char column names	[5G_FWA_Sub_Mn] — square brackets

12 PB Analysis Queries

- All 12 BigQuery queries converted to Fabric T-SQL
- The Trend_analysis join table was replaced with the REPORTING_PERIOD view
- Queries saved using a PB_ prefix naming convention
- Q1 additionally saved as a view, to drive the Power BI ranking visual

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
StagingLakehouseForDataflows_202606161		Lakehouse	—	sai verna	—	—	—	—	
StagingLakehouseForDataflows_202606162		SQL analyti...	—	sai verna	—	—	—	—	
StagingWarehouseForDataflows_20260616		Warehouse	—	sai verna	—	—	—	—	
Telecom_Dataflows	✓	Dataflow G...	—	sai verna	6/18/2026, 4:2...	—	—	—	
Telecom_Lakehouse		Lakehouse	—	sai verna	—	—	—	—	
Telecom_Lakehouse		SQL analyti...	—	sai verna	—	—	—	—	
Telecom_Report		Report	—	Telecom_pr...	6/25/2026, 3:5...	—	—	—	No
Telecom_Semantic_Model		Semantic m...	—	Telecom_pr...	6/25/2026, 3:...	N/A	—	—	
Telecom_Warehouse		Warehouse	—	sai verna	—	—	—	—	

Fig — Telecom_project workspace in Microsoft Fabric: Staging Lakehouse/Warehouse for the Dataflow run, Telecom_Lakehouse, Telecom_Dataflows (Dataflow Gen2), Telecom_Warehouse, the Telecom_Semantic_Model and the final Telecom_Report.

PHASE 4 — Power BI Dashboard Build

Connection & Architecture

- Power BI Service (cloud) connected directly to Telecom_Warehouse
- Semantic model built on top of the 12 Warehouse views — not the raw table
- Single-page layout: a fixed KPI/navigation area on top, and a dynamic visual canvas below, controlled by bookmarks
- Full build done in Microsoft Fabric cloud — no Power BI Desktop used

Bookmark-Based Navigation

Instead of separate report pages, the dashboard uses Power BI bookmarks on a single canvas. Each bookmark button on the right-hand panel swaps the visual(s) shown in the main canvas for that business question, while the KPI cards stay fixed at the top. The default landing view (before any bookmark is clicked) shows the Subscriber Growth bar chart and the Private vs PSU Market Share chart — the same Subscriber Growth chart stays visible at the top throughout, across every bookmark shown below.

Which operator captured the most new subscribers over the year?

Jio led with 26.57M net adds, followed closely by Airtel at 24.87M, together dominating overall market growth.

Did Airtel close the gap on Jio?

Airtel showed strong growth but did not overtake Jio — the gap remains, with Jio still slightly pulling ahead.

KPI Cards

Built as static shapes rather than live DAX measures — in the Warehouse/DirectLake setup, a standard-deviation-based volatility measure wasn't straightforward to build natively in Power BI Service, so these values were computed from the BigQuery/Fabric SQL outputs and entered as fixed shapes.

KPI Card	Value	Source
Growth Volatility (Std Dev)	9.00	Std dev. of monthly net subscriber additions (Q10)
Avg M2M Monthly Growth	1.86%	Average MoM M2M connection growth rate (Q7)

Tooltips

A custom tooltip page was built for every visual. Hovering over any chart pops up a card with short Q&A insights for that visual, pulled directly from the corresponding SQL analysis. Each bookmark below is shown as a pair — the dashboard view on the left, the tooltip in action on the right.

Bookmark 1 — Urban_Rural — Urban vs Rural Tele-Density Gap

How did the urban-rural gap change?

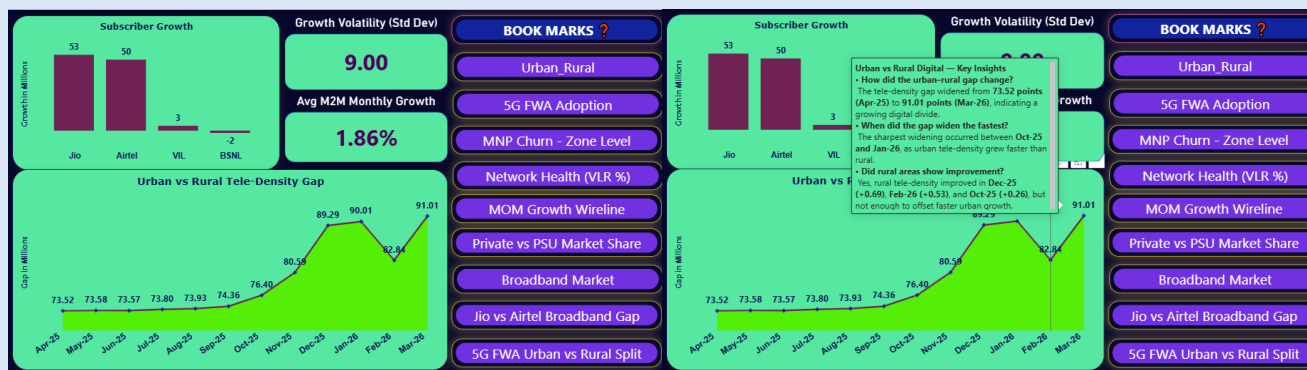
The tele-density gap widened from 73.52 points (Apr-25) to 91.01 points (Mar-26), indicating a growing digital divide.

When did the gap widen the fastest?

The sharpest widening occurred between Oct-25 and Jan-26, as urban tele-density grew faster than rural.

Did rural areas show any improvement?

Yes — rural tele-density improved in Dec-25, Feb-26 and Oct-25, but not enough to offset faster urban growth.



Dashboard view

Tooltip view

Bookmark 2 — 5G FWA Adoption Velocity

Which phase shows the strongest growth?

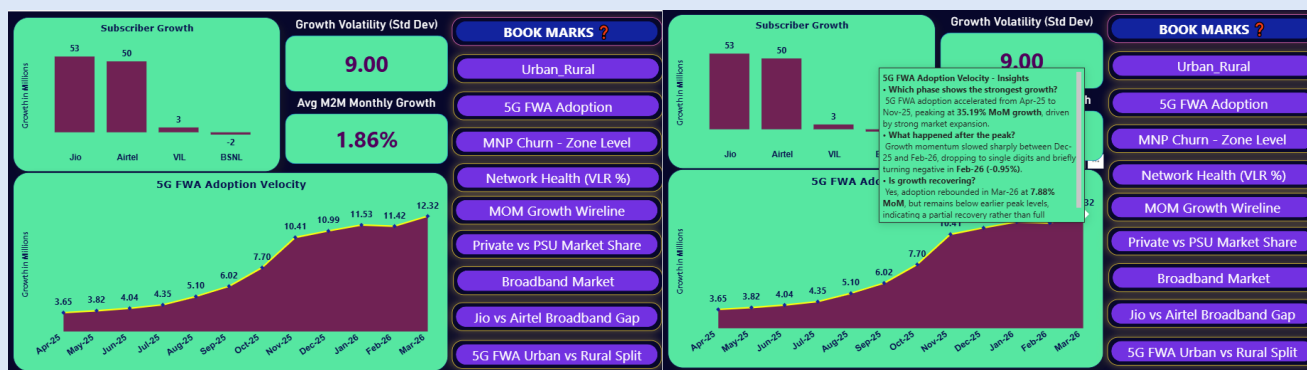
Adoption accelerated from Apr-25 to Nov-25, peaking at 35.19% MoM growth on strong market expansion.

What happened after the peak?

Momentum slowed sharply between Dec-25 and Feb-26, dropping to single digits and briefly turning negative in Feb-26 (-0.95%).

Is growth recovering?

Yes — adoption rebounded to 7.88% MoM in Mar-26, though still below the earlier peak: a partial recovery.



Dashboard view

Tooltip view

Bookmark 3 — MNP & Churn Trends — Zone Level

Which zone drives higher churn?

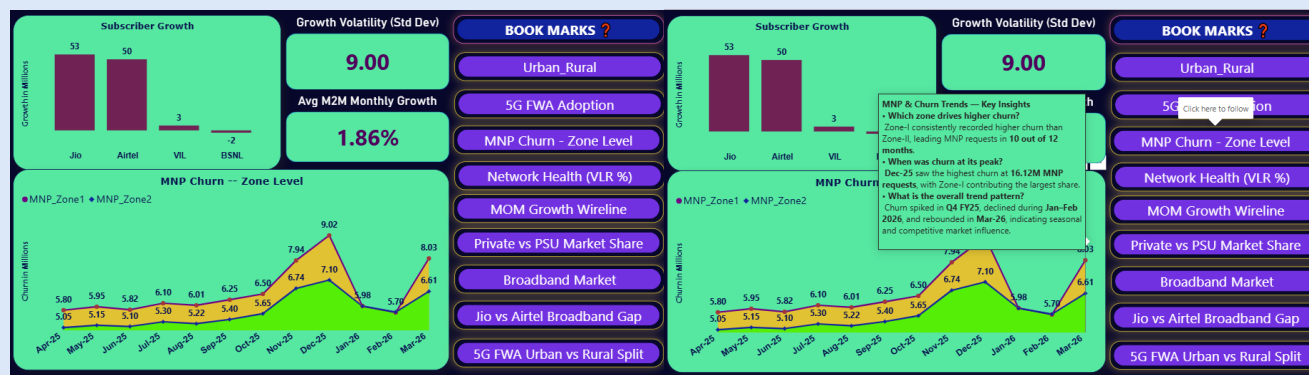
Zone-I consistently recorded higher churn than Zone-II, leading MNP requests in 10 out of the 12 months.

When was churn at its peak?

Dec-25 saw the highest churn at 16.12M MNP requests, with Zone-I contributing the largest share.

What's the overall pattern?

Churn spiked in Q4 FY25, declined through Jan–Feb 2026, then rebounded in Mar-26 — a seasonal and competitive pattern.



Bookmark 4 — Network Health — Active SIM Ratio (VLR %)

How stable is network activity?

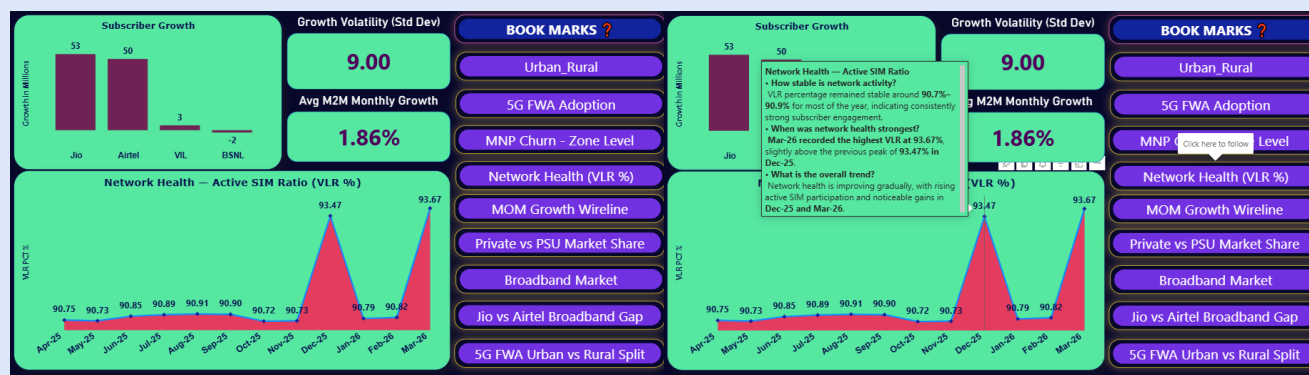
VLR % stayed stable around 90.7–90.9% for most of the year, signalling consistently strong subscriber engagement.

When was network health strongest?

Mar-26 recorded the highest VLR at 93.67%, just above the previous peak of 93.47% in Dec-25.

What's the overall trend?

Gradually improving, with rising active-SIM participation and clear gains in Dec-25 and Mar-26.



Bookmark 5 — MoM Growth — Wireline

How is the wireline segment performing overall?

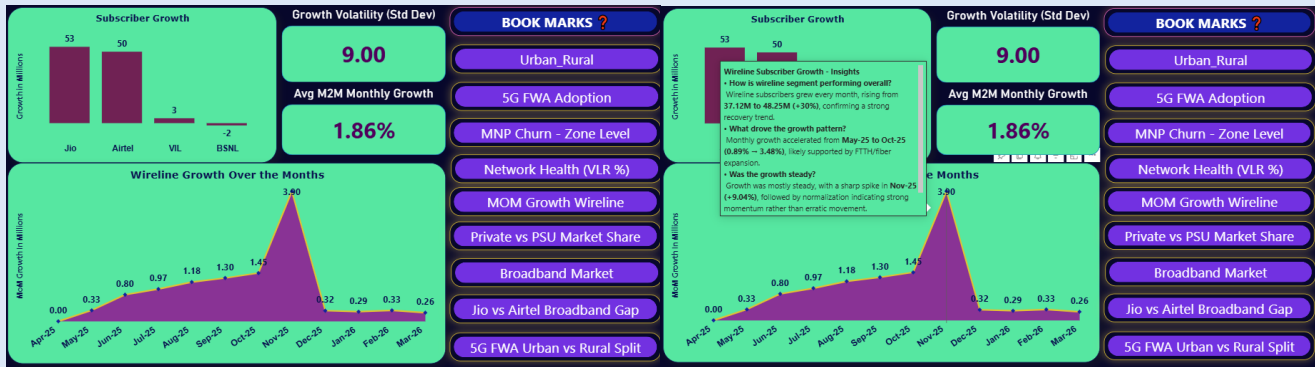
Wireline subscribers grew every month, rising from 37.12M to 48.25M (+30%) — a strong recovery trend.

What drove the growth pattern?

Monthly growth accelerated from May-25 to Oct-25 (0.89% → 3.48%), likely supported by FTTH/fibre expansion.

Was the growth steady?

Mostly steady, with a sharp spike in Nov-25 (+9.04%) followed by normalisation — momentum rather than erratic movement.



Dashboard view

Tooltip view

Bookmark 6 — Private vs PSU Market Share

This combination — Subscriber Growth on top, Private vs PSU Market Share below — is also what loads by default before any bookmark is clicked.

How is the competitive landscape changing?

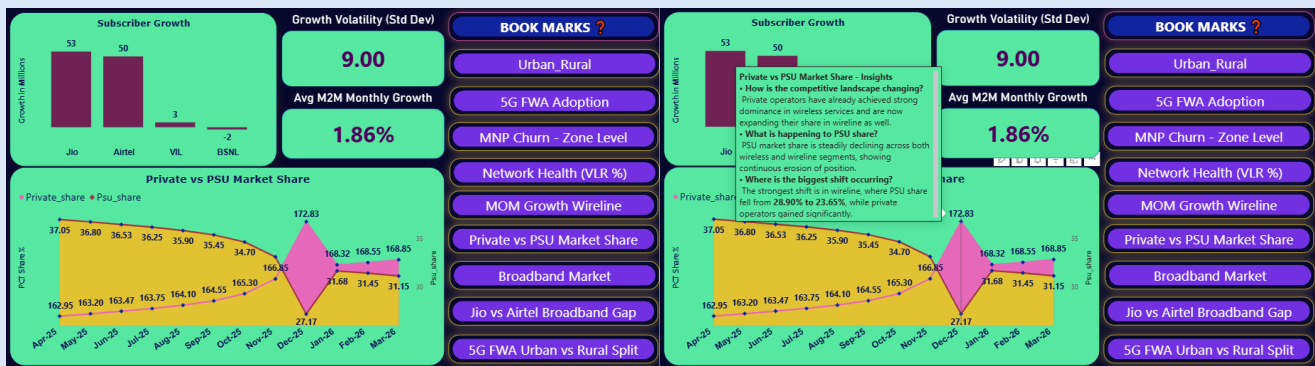
Private operators already dominate wireless and are now expanding share in wireline too.

What's happening to PSU share?

PSU market share is steadily declining across both wireless and wireline segments — continuous erosion of position.

Where is the biggest shift occurring?

The strongest shift is in wireline, where PSU share fell from 28.90% to 23.65%, while private operators gained significantly.



Dashboard view

Tooltip view

Bookmark 7 — Broadband Market Composition

How dominant is mobile broadband?

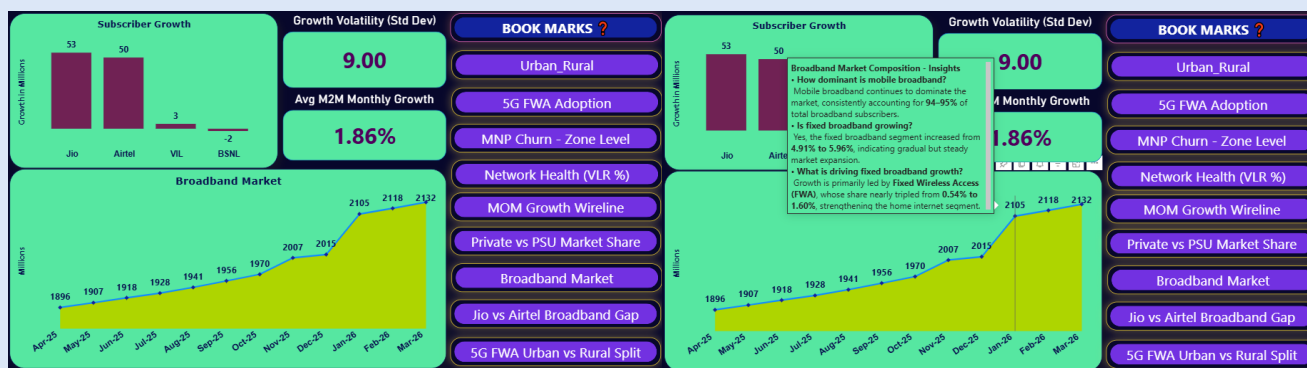
Mobile broadband continues to dominate, consistently accounting for 94–95% of total broadband subscribers.

Is fixed broadband growing?

Yes — the fixed broadband segment grew from 4.91% to 5.96%, a gradual but steady expansion.

What's driving fixed broadband growth?

Mainly Fixed Wireless Access (FWA), whose share nearly tripled from 0.54% to 1.60%, strengthening home internet.



Dashboard view



Tooltip view

Bookmark 8 — Jio vs Airtel Broadband Gap

Who is leading the market?

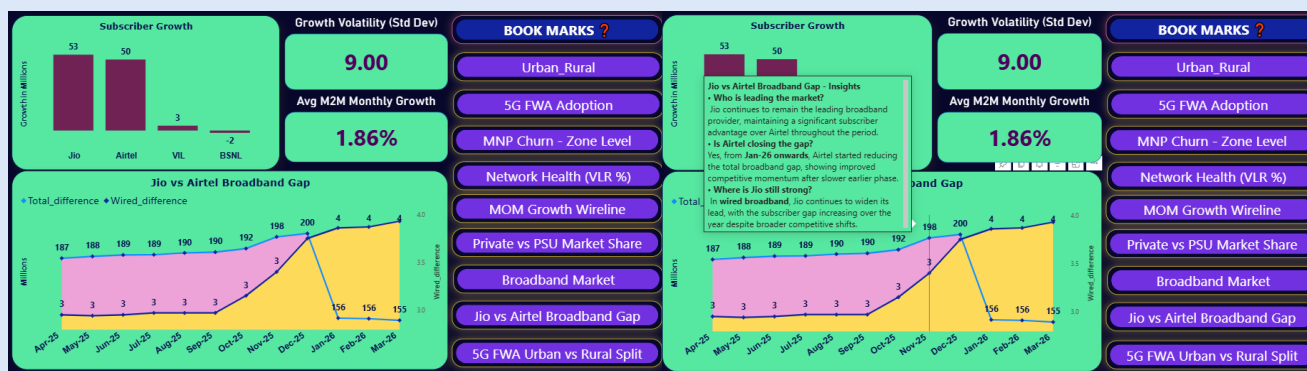
Jio remains the leading broadband provider, holding a significant subscriber advantage over Airtel throughout the period.

Is Airtel closing the gap?

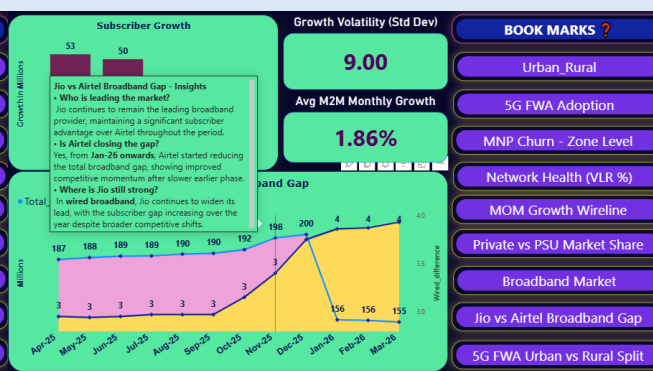
From Jan-26 onwards, Airtel started reducing the total broadband gap — improved competitive momentum after a slower start.

Where is Jio still strongest?

In wired broadband, where Jio's lead actually widens over the year despite the broader competitive shifts.



Dashboard view



Tooltip view

Bookmark 9 — 5G FWA Urban vs Rural Split

How is rural adoption trending?

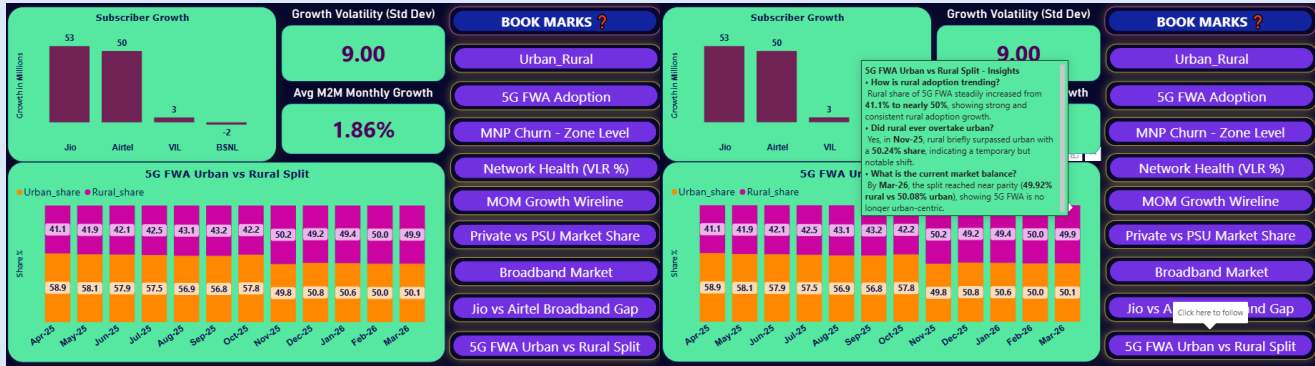
Rural share of 5G FWA rose steadily from 41.1% to nearly 50%, showing strong and consistent rural adoption growth.

Did rural ever overtake urban?

Yes — in Nov-25, rural briefly surpassed urban with a 50.24% share, a temporary but notable shift.

What's the current market balance?

By Mar-26 the split reached near parity (49.92% rural vs 50.08% urban) — 5G FWA is no longer urban-centric.



Dashboard view

Tooltip view

Key Takeaways

- Jio and Airtel together captured almost all net new subscribers (26.57M + 24.87M), leaving Vi (~1.7M) and BSNL (net losses) further behind in the competitive race.
- Private operators aren't just dominant in wireless anymore — they're now actively eating into PSU wireline share too (28.90% → 23.65%), a continuing erosion of PSU's position.
- 5G FWA adoption cooled sharply after Nov-25, but it's no longer an urban-only story — rural share is nearly at parity with urban (49.92% vs 50.08%) by Mar-26.
- The urban–rural tele-density gap kept widening (73.52 → 91.01) even though rural tele-density itself improved in several months — urban growth simply outpaced it.
- Network health (VLR %) and wireline subscriber growth both trended positively across the year, pointing to healthier overall engagement alongside the broader digital divide.

Project Flow at a Glance

